

Contents

1	Introduction	2
1.1	Probability Spaces	2
1.1.1	Exersice	5
1.2	Distributions	6
1.3	Random Variables	9
1.4	Expected Value	12
1.4.1	Inequalities	12
1.4.2	Integration to the Limit	13
1.4.3	Computing Expected Values	13
2	Law of Large Number	15
2.1	Independence	15
2.1.1	Sufficient Conditions	18
2.1.2	Independence, Distribution, and Expectation	21
2.1.3	Sums of Independent Random Variables	21
2.2	Weak Laws of Large Numbers	24
	Index	24

Introduction

1.1 Probability Spaces

Definition 1.1.1 ► Probability Spaces

A **probability space** is a measure space $(\Omega, \mathcal{F}, P)$ with

$$P : \mathcal{F} \rightarrow [0, 1]$$

Example 1.1.2 ► Discrete probability spaces

Let Ω be a countable set . Let $\mathcal{F}$ be the collection of all subsets of Ω . Let

$$P(A) = \sum_{\omega \in A} p(\omega), \quad \text{where } p(\omega) \geq 0 \text{ and } \sum_{\omega \in \Omega} p(\omega) = 1$$

Example 1.1.3 ► Measures on the real line.

Measures on $(\mathbf{R}, \mathcal{R})$ are defined by giving probability a **Stieltjes measure function** with the following properties:

- (i) F is nondecreasing.
- (ii) F is right continuous, that is, $\lim_{y \downarrow x} F(y) = F(x)$.

Theorem 1.1.4

Associated with each Stieltjes measure function F there is a unique measure μ on $(\mathbf{R}, \mathcal{R})$ with $\mu((a, b]) = F(b) - F(a)$

$$\mu((a, b]) = F(b) - F(a) \tag{1.1}$$

When $F(x) = x$ the resulting measure is called **Lebesgue measure**

Remark.

The proof is long and winding. We will show the main idea following.

Definition 1.1.5 ▶ Semialgebra

A collection $\mathcal{S}$ of sets is said to be a **semialgebra** if (i) it is closed under intersection, that is, $S, T \in \mathcal{S}$ implies $S \cap T \in \mathcal{S}$, and (ii) if $S \in \mathcal{S}$ then S^c is a finite disjoint union of sets in $\mathcal{S}$.

Example 1.1.6

$S_d =$ the empty set plus all sets of the form

$$(a_1, b_1] \times \cdots \times (a_d, b_d] \subset \mathbb{R}^d \quad \text{where } -\infty \leq a_i < b_i \leq \infty$$

Then S_d is a semialgebra.

Remark.

The above Stieljes measure function gives the values of μ on the semialgebra.

Definition 1.1.7 ▶ Algebra

A collection $\mathcal{A}$ of subsets of Ω is called an **algebra** (or **field**) if $A, B \in \mathcal{A}$ implies A^c and $A \cup B$ are in $\mathcal{A}$.

Remark.

Since $A \cap B = (A^c \cup B^c)^c$, it follows that $A \cap B \in \mathcal{A}$. Obviously a σ -algebra is an algebra.

Lemma 1.1.8

If $\mathcal{S}$ is a semialgebra, then $\bar{\mathcal{S}} = \{\text{finite disjoint unions of sets in } \mathcal{S}\}$ is an algebra, called the **algebra generated by $\mathcal{S}$** .

Remark.

Given a set function μ on $\mathcal{S}$, we can extend it to $\bar{\mathcal{S}}$ by

$$\mu\left(+_{i=1}^n A_i\right) = \sum_{i=1}^n \mu(A_i)$$

Proof. Suppose $A = +_i S_i$ and $B = +_j T_j$, where $+$ denotes disjoint union and we assume the index sets are finite. Then $A \cup B = +_{i,j} S_i \cap T_j$, where $S_j \cap T_j \in \mathcal{S}$

by definition. Thus $\bar{\mathcal{S}}$ is closed under intersection. As for complements, if $A = +_i \mathcal{S}_i$, then $A^c = \cap_i \mathcal{S}_i^c$, where $\mathcal{S}_i^c$ is a finite disjoint union of sets in $\mathcal{S}$, thus is in $\bar{\mathcal{S}}$. Since $\bar{\mathcal{S}}$ is closed under intersection, then $A^c \in \bar{\mathcal{S}}$ by induction. $\square$

Definition 1.1.9 ► measure on an algebra

By a **measure on an algebra** $\mathcal{A}$, we mean a set function μ with

- (i) $\mu(A) \geq \mu(\emptyset) = 0$ for all $A \in \mathcal{A}$, and
- (ii) if $A_i \in \mathcal{A}$ are **disjoint** and **their union is in** $\mathcal{A}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Definition 1.1.10 ► σ -finite

μ is said to be **σ -finite** if there is a sequence of sets $A_n \in \mathcal{A}$ so that $\mu(A_n) < \infty$ and $\cup_n A_n = \Omega$.

Remark.

Letting $A'_1 = A_1$ and for $n \geq 2$,

$$A'_n = \bigcup_{m=1}^n A_m \quad \text{or} \quad A'_n = A_n \cap \left(\bigcap_{m=1}^{n-1} A_m^c \right) \in \mathcal{A}$$

we can without loss of generality assume that $A_n \uparrow \Omega$ or the A_n are disjoint.

Theorem 1.1.11

Let $\mathcal{S}$ be a semialgebra and let μ defined on $\mathcal{S}$ have $\mu(\emptyset) = 0$. Suppose (i) if $S \in \mathcal{S}$ is a finite disjoint union of sets $S_i \in \mathcal{S}$ then $\mu(S) = \sum_i \mu(S_i)$, and (ii) if $S_i, S \in \mathcal{S}$ with $S = +_{i \geq 1} S_i$ then $\mu(S) \leq \sum_{i \geq 1} \mu(S_i)$. Then μ has a unique extension $\bar{\mu}$ that is a measure on $\bar{\mathcal{S}}$ the algebra generated by $\mathcal{S}$. If $\bar{\mu}$ is sigma-finite then there is a unique extension ν that is a measure on $\sigma(\mathcal{S})$.

Lemma 1.1.12

Suppose only that (i) holds.

1. If $A, B_i \in \bar{\mathcal{S}}$ with $A = +_{i=1}^n B_i$, then $\bar{\mu}(A) = \sum_i \bar{\mu}(B_i)$.

2. If $A, B_i \in \bar{\mathcal{S}}$ with $A \subseteq \bigcup_{i=1}^n B_i$ then $\bar{\mu}(A) \leq \sum_i \bar{\mu}(B_i)$

Measures on $\mathbb{R}^d$

1.1.1 Exercice

Exercise 1.1.13

A set $A \subset \{1, 2, \dots\}$ is said to have *asymptotic density* θ if

$$\lim_{n \rightarrow \infty} |A \cap \{1, 2, \dots, n\}|/n = \theta$$

Let $\mathcal{A}$ be the collection of sets for which the asymptotic density exists. Is $\mathcal{A}$ a σ -algebra? an algebra?

Proof. Suppose that $A \in \mathcal{A}$ has asymptotic density θ .

$$|A \cap \{1, 2, \dots, n\}|/n + |A^c \cap \{1, 2, \dots, n\}|/n = 1$$

Take the limit of $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} |A \cap \{1, 2, \dots, n\}|/n + \lim_{n \rightarrow \infty} |A^c \cap \{1, \dots, n\}|/n = 1$$

which implies that A^c has the asymptotic density $1 - \theta$.

Let $A = \{2k\}_{k=1}^{\infty}$. Let B contained all the even numbers among $[2^{2m}, 2^{2m+1} - 1]$, and contained all the odd numbers among $[2^{2m+1}, 2^{2m+2} - 1]$. Since there are as many even numbers as odd numbers among $[2^k, 2^k - 1]$. It is not hard to see B has asymptotic density $\frac{1}{2}$. Then

$$\begin{aligned} \lim_{m \rightarrow \infty} |(A \cap B) \cap \{1, 2, \dots, 2^{2m+1}\}|/2^{2m+1} &= \lim_{m \rightarrow \infty} \sum_{k=0}^m \frac{1}{2} (2^{2k+1} - 2^{2k})/2^{2m+1} \\ &= \lim_{m \rightarrow \infty} \sum_{k=0}^m 4^{k-m-1} \\ &= \lim_{m \rightarrow \infty} 4^{-m-1} \frac{1 - 4^{m+1}}{1 - 4} = \frac{1}{3} \end{aligned}$$

$$\lim_{m \rightarrow \infty} |(A \cap B) \cap \{1, 2, \dots, 2^{2m+2}\}| / 2^{2m+2} = \lim_{m \rightarrow \infty} \sum_{k=0}^m \frac{1}{2} (2^{2m+1} - 2^{2m}) / 2^{2m+2} = \frac{1}{6}$$

which implies that $A \cap B$ is not in $\mathcal{A}$. But $\mathcal{A}$ is closed under complements, $\mathcal{A}$ is not closed under finite union. Thus $\mathcal{A}$ is not an algebra, then is neither a σ -algebra. $\square$

1.2 Distributions

Definition 1.2.1

A real-valued function X defined on Ω is said to be a **random variable** if for every Borel set $B \subseteq \mathbb{R}$ we have $X^{-1}(B) \in \mathcal{F}$. That is a real valued measurable function on $\mathcal{F}$. We write $X \in \mathcal{F}$.

Example 1.2.2

1. If Ω is a discrete probability space, then any function $X : \Omega \rightarrow \mathbb{R}$ is a random variable.
2. For any $A \in \mathcal{F}$, the **indicator function** of A is:

$$1_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

Definition 1.2.3 ► distribution

- If X is a random variable, then X induces a probability measure on $\mathbb{R}$ called **distribution** by setting $\mu(A) = P(X \in A)$ for Borel sets A .
- The distribution of a random variable X is usually described by giving its **distribution function**,

$$F(x) = P(X \leq x)$$

Remark.

The right-hand side can be written as $P(X^{-1}(A))$. In words, we pull $A \in \mathbb{R}$ back to $X^{-1}(A) \in \mathcal{F}$ and then take P of that set.

Proof. If the A_i are disjoint,

$$\begin{aligned}\mu\left(\bigcup_i A_i\right) &= P(X \in A) \\ &= P\left(\bigcup_i \{X \in A_i\}\right) \\ &= \sum_i P(X \in A_i) = \sum_i \mu(A_i)\end{aligned}$$

□

Theorem 1.2.4

Any distribution function F has the following properties:

- (i) F is **nondecreasing**.
- (ii) $\lim_{x \rightarrow \infty} F(x) = 1$, $\lim_{x \rightarrow -\infty} F(x) = 0$.
- (iii) F is **right continuous**, that is, $\lim_{y \downarrow x} F(y) = F(x)$.
- (iv) If $F(x-) = \lim_{y \uparrow x} F(y)$ then $F(x-) = P(X < x)$.
- (v) $P(X = x) = F(x) - F(x-)$.

Theorem 1.2.5

If F satisfies (i), (ii), and (iii) in the above theorem, then it is the distribution function of some random variable.

Remark.

Even though F may not be 1-1 and onto, we will call X the inverse of F and denote by F^{-1} .

Proof. Let P be Lebesgue measure, Note that

$$F(x) = P(\omega : \omega \leq F(x))$$

And if X is a random variable which we need, then

$$F(x) = P(\omega : X(\omega) \leq x)$$

Thus we only need to show that

$$\{\omega : X(\omega) \leq x\} = \{\omega : \omega \leq F(x)\}$$

holds for some X . If F is a bijection, we may define

$$X(\omega) = F^{-1}(\omega)$$

It motivates us to define

$$X(\omega) = \sup \{y : F(y) < \omega\}$$

in a general case. To check that it is just what we need, we observe that if $\omega \leq F(x)$, and if $x \leq X(\omega)$, then $F(x) < \omega$, which is a contradiction. Thus $x > X(\omega)$. On the other hand if $\omega > F(x)$, then since F is right continuous, there is an $\varepsilon > 0$ so that $F(x + \varepsilon) < \omega$, and $X(\omega) \geq x + \varepsilon > x$. $\square$

Definition 1.2.6 ► Equal in Distribution

- If X and Y induce the same distribution μ on $(\mathbb{R}, \mathcal{R})$, we say X and Y are **equal in distribution**, written by

$$X =_d Y$$

- When the distribution function $F(x) = P(X \leq x)$ has the form

$$F(x) = \int_{-\infty}^x f(y) \, dy$$

We say that X has **density function** f . We write $f_X(x)$.

Remark.

- X and Y are equal in distribution, iff X and Y have the same distribution function, that is, $P(X \leq x) = P(Y \leq x)$ for all x .
- In remembering formulas, it is often useful to think of $f(x)$ as being $P(X = x)$ although

$$P(X = x) = F(x) - F(x-) = \lim_{\varepsilon \rightarrow 0} \int_{x-\varepsilon}^{x+\varepsilon} f(y) \, dy = 0$$

Example 1.2.7 ▶ Uniform distribution on (0,1)

$f(x) = 1$ for $x \in (0, 1)$ and 0 otherwise. Distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 \leq x \leq 1 \\ 1 & x > 1 \end{cases}$$

Example 1.2.8 ▶ Exponential distribution with rate λ

$f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and 0 otherwise. Distribution function:

$$F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

Example 1.2.9 ▶ Standard normal distribution

$$f(x) = (2\pi)^{-1/2} \exp(-x^2/2)$$

Remark.

In this case, there is no closed-form expression for $F(x)$, but we have the following bounds that are useful for large x :

Theorem 1.2.10 ▶ For $x > 0$

$$(x^{-1} - x^{-3}) \exp(-x^2/2) \leq \int_x^\infty \exp(-y^2/2) dy \leq x^{-1} \exp(-x^2/2)$$

1.3 Random Variables

Definition 1.3.1

A function $X : \Omega \rightarrow S$ is said to be a **measurable map** from $(\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ if

$$X^{-1}(B) \equiv \{\omega : X(\omega) \in B\} \in \mathcal{F} \quad \text{for all } B \in \mathcal{S}$$

If $(S, \mathcal{S}) = (\mathbb{R}^d, \mathcal{R}^d)$ and $d > 1$, then X is called a **random vector**. If $d = 1$, X is called a **random variable**, ro r.v. for short.

Theorem 1.3.2

If $\{\omega : X(\omega) \in A\} \in \mathcal{F}$ for all $A \in \mathcal{A}$ and $\mathcal{A}$ **generates** $\mathcal{S}$, then X is measurable.

Definition 1.3.3

Suppose that X is a measurable map. Then we define

$$\sigma(X) := \{\{X \in B\} : B \in \mathcal{S}\}$$

as the **σ -field generated by X** .

Remark.

where we write $\{X \in B\}$ as shorthand for $\{\omega : X(\omega) \in B\}$.

Example 1.3.4

If $(S, \mathcal{S}) = (\mathbb{R}, \mathcal{R})$, then possible choices of $\mathcal{A}$ in 1.3.3 are $\{(-\infty, x] : x \in \mathbb{R}\}$ or $\{(-\infty, x) : x \in \mathbb{Q}\}$.

Example 1.3.5

If $(S, \mathcal{S}) = (\mathbb{R}^d, \mathcal{R}^d)$, a useful choice of $\mathcal{A}$ is

$$\{(a_1, b_1) \times \cdots \times (a_d, b_d) : -\infty < a_i < b_i < \infty\}$$

Theorem 1.3.6

If $X : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{S})$ and $f : (S, \mathcal{S}) \rightarrow (T, \mathcal{T})$ are measurable maps, then $f(X)$ is a measurable map from $(\Omega, \mathcal{F})$ to $(T, \mathcal{T})$.

Theorem 1.3.7

If $X_1, \dots, X_n$ are random variables and $f : (\mathbb{R}^n, \mathcal{R}^n) \rightarrow (\mathbb{R}, \mathcal{R})$ is measurable, then $f(X_1, \dots, X_n)$ is a random variable.

Theorem 1.3.8

If $X_1, \dots, X_n$ are random variables then $X_1 + \cdots + X_n$ is a random variable.

Theorem 1.3.9

f $X_1, X_2, \dots$ are random variables then so are

$$\inf_n X_n \quad \sup_n X_n \quad \limsup_n X_n \quad \liminf_n X_n$$

Exercise**Exercise 1.3.10**

Show that if $\mathcal{A}$ generates $\mathcal{S}$, then $X^{-1}(\mathcal{A}) \equiv \{\{X \in A\} : A \in \mathcal{A}\}$ generates $\sigma(X) = \{\{X \in B\} : B \in \mathcal{S}\}$

Proof. First, we show that

$$\sigma(X^{-1}(\mathcal{A})) \subseteq X^{-1}(\mathcal{S})$$

For which, we only need to show that

1. $X^{-1}(\mathcal{S})$ is a σ -algebra;
 2. $X^{-1}(\mathcal{A}) \subseteq X^{-1}(\mathcal{S})$.
2. is obvious from $\mathcal{A} \subseteq \mathcal{S}$. Since $\mathcal{S}$ is a σ -algebra, and the complementation, union are kept under preimage, then $X^{-1}(\mathcal{S})$ is a σ -algebra.

Next, we show that

$$X^{-1}(\mathcal{S}) \subseteq \sigma(X^{-1}(\mathcal{A}))$$

We define

$$\mathcal{C} := \{B \in \mathcal{S} : X^{-1}(B) \in \sigma(X^{-1}(\mathcal{A}))\}$$

It is not hard to see that $\mathcal{C}$ is a σ -algebra. Once we show that $\mathcal{A} \subseteq \mathcal{C}$, we have $\mathcal{S} \subseteq \mathcal{C}$, then it follows that $X^{-1}(\mathcal{S}) \subseteq \sigma(X^{-1}(\mathcal{A}))$. For that, we take $A \in \mathcal{A}$, since $X^{-1}(A) \in X^{-1}(\mathcal{A}) \subseteq \sigma(X^{-1}(\mathcal{A}))$, $A \in \mathcal{C}$. Thus we completes the proof. $\square$

1.4 Expected Value

Definition 1.4.1 ► Expected Value

- If $X \geq 0$ is a random variable on $(\Omega, \mathcal{F}, P)$, then we define its **expected value** to be

$$EX = \int X \, dP$$

We declare that EX **exists** and set

$$EX = EX^+ - EX^-$$

whenever the subtraction makes sense.

Remark.

EX is often called the **mean** of X and denoted by μ .

Theorem 1.4.2

Suppose $X, Y \geq 0$ or $E|X|, E|Y| < \infty$.

- $E(X + Y) = EX + EY$.
- $E(aX + b) = aE(X) + b$ for any real numbers a, b .
- If $X \geq Y$ then $EX \geq EY$.

1.4.1 Inequalities

Theorem 1.4.3 ► Jensen

Suppose φ is **convex**. Then

$$E(\varphi(X)) \geq \varphi(EX)$$

provided both expectations exist, that is, $E|X|$ and $E|\varphi(X)| < \infty$

Theorem 1.4.4 ► Holder

If $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$, then

$$E|XY| \leq \|X\|_p \|Y\|_q$$

Here $\|X\|_r = (E|X|^r)^{\frac{1}{r}}$ for $r \in [1, \infty)$; $\|X\|_\infty = \inf\{M : P(|X| > M) = 0\}$

1.4.2 Integration to the Limit

Theorem 1.4.5 ▶ Fatou's lemma

If $X_n \geq 0$ then

$$\liminf_{n \rightarrow \infty} EX_n \geq E\left(\liminf_{n \rightarrow \infty} X_n\right)$$

Theorem 1.4.6 ▶ Monotone convergence theorem

If $0 \leq X_n \uparrow X$ then $EX_n \uparrow EX$.

Theorem 1.4.7 ▶ Dominated convergence theorem

If $X_n \rightarrow X$ a.s., $|X_n| \leq Y$ for all n , and $EY < \infty$, then $EX_n \rightarrow EX$.

1.4.3 Computing Expected Values

Theorem 1.4.8 ▶ Change of variables formula

Let X be a random element of $(S, \mathcal{S})$ with distribution μ , that is, $\mu(A) = P(X \in A)$. If f is a measurable function from $(S, \mathcal{S})$ to $(\mathbb{R}, \mathcal{R})$ so that $f \geq 0$ or $E|f(X)| < \infty$, then

$$Ef(X) = \int_S f \, d\mu =: \int_S f(y) \mu(dy)$$

Definition 1.4.9

- If k is a positive integer, then EX^k is called the ***kth moment*** of X . The first moment EX is usually called the ***mean*** and denoted by μ .
- If $EX^2 < \infty$, then the ***variance*** of X is defined to be

$$\begin{aligned} \text{var}(X) &= E(X - \mu)^2 \\ &= EX^2 - \mu^2 \end{aligned}$$

Proposition 1.4.10

- $\text{var}(X) \leq EX^2$
- $\text{var}(aX + b) = a^2 \text{var}(X)$

Example 1.4.11

If X has the *exponential distribution* with rate 1, then

$$EX^k = \int_0^{\infty} x^k e^{-x} dx = k!$$

Proof.

$$\begin{aligned} EX^k &= \int_S y^k d\mu = \int_S y^k dF = \int_{\mathbb{R}} x^k f(x) dx = \int_0^{\infty} x^k e^{-x} dx \\ \int_0^{\infty} x^k e^{-x} dx &= - \int_0^{\infty} x^k de^{-x} = -[e^{-x} x^k]_0^{\infty} + k \int_0^{\infty} x^{k-1} e^{-x} dx \\ &= k \int_0^{\infty} x^{k-1} e^{-x} dx \end{aligned}$$

Then it follows from induction. □

Law of Large Number

2.1 Independence

Definition 2.1.1 ► Independence

- Two events A and B are **independent** if $P(A \cap B) = P(A)P(B)$.
- Two random variables X and Y are **independent** if for all $C, D \in \mathcal{R}$,

$$P(X \in C, Y \in D) = P(X \in C)P(Y \in D)$$

that is, the events $A = \{X \in C\}$ and $B = \{Y \in D\}$ are independent.

- Two σ -fields $\mathcal{F}$ and $\mathcal{G}$ are **independent** if for all $A \in \mathcal{F}$ and $B \in \mathcal{G}$ the events A and B are independent.

Exercise 2.1.2

1. Show that if A and B are independent, then so are A^c and B , A and B^c , and A^c and B^c .
2. Conclude that events A and B are independent if and only if their indicator random variables 1_A and 1_B are independent.

Proof. 1. If A and B are independent, then

$$P(A \cap B) = P(A)P(B)$$

Note that

$$P(A \cap B) + P(A^c \cap B) = P(B)$$

We have

$$P(A^c \cap B) = P(B)(1 - P(A)) = P(B)P(A^c)$$

Similarly,

$$P(A \cap B^c) = P(A)P(B^c)$$

$$P(A^c \cap B^c) = P(B^c) - P(A \cap B^c) = P(B^c) - P(A)P(B^c) = P(A^c)P(B^c)$$

2.

$$\sigma(1_A) = \{\emptyset, A, A^c, \Omega\}, \quad \sigma(1_B) = \{\emptyset, B, B^c, \Omega\}$$

If $\sigma(1_A)$ and $\sigma(1_B)$ are independent, it is clear that A and B are as well. Conversely, only need to show that $\emptyset$ and A are independent, the remaining follows from 1. and symmetry. We have

$$P(\emptyset \cap A) = 0 = P(\emptyset)P(A)$$

□

Definition 2.1.3

- σ -fields $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n$ are **independent** if whenever $A_i \in \mathcal{F}_i$ for $i = 1, \dots, n$, we have

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i)$$

- Random variables $X_1, \dots, X_n$ are **independent** if whenever $B_i \in \mathcal{R}$ for $i = 1, \dots, n$ we have

$$P\left(\bigcap_{i=1}^n \{X_i \in B_i\}\right) = \prod_{i=1}^n P(X_i \in B_i)$$

- Sets $A_1, \dots, A_n$ are **independent** if whenever $I \subset \{1, \dots, n\}$ we have

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

Remark.

The last definition looks slightly different from the former two, but it is consistent with the independence of the indicator variables 1_{A_i} , $1 \leq i \leq n$.

Exercise 2.1.4

Let $A_1, A_2, \dots, A_n$ be independent. Show

1. $A_1^c, A_2, \dots, A_n$ are independent;
2. $1_{A_1}, \dots, 1_{A_n}$ are independent.

Proof. 1. If $1 \notin I$, then it is obvious that

$$P(\cap_{i \in I} A_i) = \prod_{i \in I} P(A_i).$$

If $1 \in I$, note that

$$\begin{aligned} P(\cap_{i \in I \setminus \{1\}} A_i \cap A_1^c) &= P(\cap_{i \in I \setminus \{1\}} A_i) - P(\cap_{i \in I \setminus \{1\}} A_i \cap A_1) \\ &= \prod_{i \in I \setminus \{1\}} P(A_i) - \prod_{i \in I} P(A_i) \\ &= (1 - P(A_1)) \prod_{i \in I \setminus \{1\}} P(A_i) \\ &= P(A_1^c) \prod_{i \in I \setminus \{1\}} P(A_i) \end{aligned}$$

2.

$$\sigma(1_{A_i}) = \{\emptyset, A_i^c, A_i, \Omega\}$$

Take $B_i \in \sigma(1_{A_i})$, $i = 1, \dots, n$. $B_i = \emptyset$ for some i , then both $P(\cap_{i=1}^n B_i)$ and $\prod_{i=1}^n P(B_i)$ are zero. The remainings follows from 1. □

Example 2.1.5

Let X_1, X_2, X_3 be independent random variables with

$$P(X_i = 0) = P(X_i = 1) = 1/2$$

Let $A_1 = \{X_2 = X_3\}$, $A_2 = \{X_3 = X_1\}$ and $A_3 = \{X_1 = X_2\}$. These events are pairwise independent since if $i \neq j$, then

$$P(A_i \cap A_j) = P(X_1 = X_2 = X_3) = 1/4 = P(A_i)P(A_j)$$

but they are not independent since

$$P(A_1 \cap A_2 \cap A_3) = 1/4 \neq 1/8 = P(A_1)P(A_2)P(A_3)$$

2.1.1 Sufficient Conditions

Definition 2.1.6 ▶ Independence between collections

Collections of sets $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n \subset \mathcal{F}$ are said to be **independent** if whenever $A_i \in \mathcal{A}_i$ and $I \subset \{1, \dots, n\}$ we have $P(\bigcap_{i \in I} A_i) = \prod_{i \in I} P(A_i)$.

Lemma 2.1.7

Without loss of generality we can suppose each $\mathcal{A}_i$ contains Ω . In this case the condition is equivalent to

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i) \text{ whenever } A_i \in \mathcal{A}_i$$

since we can set $A_i = \Omega$ for $i \notin I$.

Definition 2.1.8 ▶ π, λ -systems

- We say that $\mathcal{A}$ is a **π -system** if it is closed under intersection, that is, if $A, B \in \mathcal{A}$ then $A \cap B \in \mathcal{A}$.
- We say that $\mathcal{L}$ is a **λ -system** if
 - (i) $\Omega \in \mathcal{L}$.
 - (ii) If $A, B \in \mathcal{L}$ and $A \subset B$, then $B - A \in \mathcal{L}$.
 - (iii) If $A_n \in \mathcal{L}$ and $A_n \uparrow A$, then $A \in \mathcal{L}$.

Theorem 2.1.9 ▶ $\pi - \lambda$ Theorem

If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a λ -system that contains $\mathcal{P}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Theorem 2.1.10

Suppose $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent and each $\mathcal{A}_i$ is a π -system. Then $\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \dots, \sigma(\mathcal{A}_n)$ are independent.

Proof sketch.

对于任一取定的事件 F , “(取定任一集族中) 与 F 独立” 的性质集是一个 λ -system. 由于 $\mathcal{A}_1$ 是 π -system, 于是 $\mathcal{A}_1$ 中与 F 独立的集合包含了 $\sigma(\mathcal{A}_1)$, 通过任意的 F , 传递归纳集族的独立性一步步“升级”到它们的生成 σ -代数上.

Remark.

It is not easy to show that if $A, B \in \mathcal{L}$ then $A \cap B \in \mathcal{L}$ or $A \cup B \in \mathcal{L}$, but it is easy to check that if $A, B \in \mathcal{L}$ with $A \subseteq B$, then $B - A \in \mathcal{L}$.

Proof. Let $F \in \mathcal{P}$ be any event. We define

$$\mathcal{L} := \{A \in \mathcal{A}_1 : P(A \cap F) = P(A)P(F)\}$$

We check that $\mathcal{L}$ is a λ -system. (i) follows from $P(A \cap \Omega) = P(A) \cdot 1 = P(A)P(\Omega)$. To show (ii), we take $A, B \in \mathcal{L}, B \supseteq A$. Note that

$$(B - A) \cap F = (B \cap F) - (A \cap F),$$

we have

$$\begin{aligned} P((B \setminus A) \cap F) &= P((B \cap F) - (A \cap F)) \\ &= P(B \cap F) - P(A \cap F) \\ &= P(B)P(F) - P(A)P(F) \\ &= (P(B) - P(A))P(F) = P((B - A))P(F) \end{aligned}$$

which shows that $(B - A) \in \mathcal{L}$

For (iii), take $B_n \in \mathcal{L}, B_n \uparrow B$. Then since $(B_n \cap F) \uparrow B \cap F$, we have

$$\begin{aligned} P(B \cap F) &= \lim_{n \rightarrow \infty} P(B_n \cap F) \\ &= \lim_{n \rightarrow \infty} P(B_n)P(F), \\ &= P(B)P(F) \end{aligned}$$

which implies that $B \in \mathcal{L}$. Thus $\mathcal{L}$ is a λ -system. It follows from $\pi - \lambda$ Theorem that $\mathcal{L} \supseteq \sigma(\mathcal{A}_1)$ For each $A_2, \dots, A_n, A_i \in \mathcal{A}_i, i = 2, 3, \dots, n$, by taking $F = \bigcap_{i=2}^n A_i$, we have

$$P(A \cap A_2 \cap \dots \cap A_n) = P(A)P\left(\bigcap_{i=2}^n A_i\right) = P(A)P(A_2) \dots P(A_n), \quad \forall A \in \sigma(\mathcal{A}_1)$$

We get that

if $\mathcal{A}_1, \dots, \mathcal{A}_n$ are independent π -systems, then $\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$ are as well (*)

By applying (*) into $\mathcal{A}_2, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$, we get $\sigma(\mathcal{A}_2), \mathcal{A}_3, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$ are

independent. The proof completes after n iterations. $\square$

Theorem 2.1.11

In order for $X_1, \dots, X_n$ to be independent, it is sufficient that for all $x_1, \dots, x_n \in (-\infty, \infty]$

$$P(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$$

Proof. Let $\mathcal{A}_i$ be the collection of the sets of the form $\{X_i \leq x_i\}$. Since

$$\{X_i \leq x\} \cap \{X_i \leq y\} = \{X_i \leq x \wedge y\}$$

$\mathcal{A}_i$ is a π -system. Since we allowed $x_i = \infty$, $\Omega \in \mathcal{A}_i$. Exercise 1.3.10 implies that $\sigma(\mathcal{A}_i) = \sigma(X_i)$, so the result follows from Theorem 2.1.11. $\square$

Theorem 2.1.12

Suppose $\mathcal{F}_{i,j}$, $1 \leq i \leq n$, $1 \leq j \leq m(i)$ are independent π -systems and let $\mathcal{G}_i = \sigma(\bigcup_j \mathcal{F}_{i,j})$. Then $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent.

Proof. Let $\mathcal{A}_i$ be the collection of sets of the form $\bigcap_j A_{i,j}$ where $A_{i,j} \in \mathcal{F}_{i,j}$. $\mathcal{A}_i$ is a π -system that contains Ω and contains $\bigcup_j \mathcal{F}_{i,j}$, so Theorem 2.1.10 implies $\sigma(\mathcal{A}_i) = \mathcal{G}_i$ are independent. $\square$

Theorem 2.1.13

If for $1 \leq i \leq n$, $1 \leq j \leq m(i)$, $X_{i,j}$ are independent and $f_i : \mathbf{R}^{m(i)} \rightarrow \mathbf{R}$ are measurable, then $f_i(X_{i,1}, \dots, X_{i,m(i)})$ are independent.

Proof. $\sigma(X_{i,j})$, $1 \leq i \leq n$, $1 \leq j \leq m(i)$ are independent π -systems. Then $\mathcal{G}_i = \sigma(\bigcup_j (\sigma(X_{i,j})))$ are independent. Take any Borel set $B \subseteq \mathbf{R}$, then

$$\begin{aligned} (f_i(X_{i,1}, \dots, X_{i,m(i)}))^{-1}(B) &= (X_{i,1}, \dots, X_{i,m(i)})^{-1}(f_i^{-1}(B)) \\ &\subseteq \sigma\left(\bigcup_j (\sigma(X_{i,j}))\right) \end{aligned}$$

Thus $f_i(X_{i,1}, \dots, X_{i,m(i)})$ are independent. □

2.1.2 Independence, Distribution, and Expectation

Theorem 2.1.14

Suppose $X_1, \dots, X_n$ are independent random variables and X_i has distribution μ_i . Then $(X_1, \dots, X_n)$ has distribution $\mu_1 \times \dots \times \mu_n$.

Theorem 2.1.15

Suppose X and Y are independent and have distributions μ and ν . If $h : \mathbf{R}^2 \rightarrow \mathbf{R}$ is a measurable function with $h \geq 0$ or $E|h(X, Y)| < \infty$, then

$$Eh(X, Y) = \iint h(x, y)\mu(dx)\nu(dy)$$

In particular, if $h(x, y) = f(x)g(y)$ where $f, g : \mathbf{R} \rightarrow \mathbf{R}$ are measurable functions with $f, g \geq 0$ or $E|f(X)|$ and $E|g(Y)| < \infty$, then

$$Ef(X)g(Y) = Ef(X) \cdot Eg(Y)$$

Theorem 2.1.16

If $X_1, \dots, X_n$ are independent and have

- (a) $X_i \geq 0$ for all i , or
- (b) $E|X_i| < \infty$ for all i , then

$$E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n EX_i$$

that is, the expectation on the left exists and has the value given on the right.

2.1.3 Sums of Independent Random Variables

Theorem 2.1.17

If X and Y are independent, $F(x) = P(X \leq x)$, and $G(y) = P(Y \leq y)$, then

$$P(X + Y \leq z) = \int F(z - y)dG(y)$$

The integral on the right-hand side is called the **convolution** of F and G and is denoted $F * G(z)$. The meaning of $dG(y)$ will be explained in the proof.

Proof.

$$\begin{aligned}
 P(X + Y \leq z) &= \int 1_{(x+y \leq z)} d(\mu \times \nu) \\
 &= \int \int 1_{(x+y \leq z)} d\mu d\nu \\
 &= \int F(z - y) d\nu \\
 &= \int F(z - y) dG(y)
 \end{aligned}$$

where $dG(y)$ is just a notation for $d\nu$. We apply this notation motivated from the Lebesgue-Stieljes integration. $\square$

Theorem 2.1.18

Suppose that X with density f and Y with distribution function G are independent. Then $X + Y$ has density

$$h(x) = \int f(x - y)dG(y)$$

When Y has density g , the last formula can be written as

$$h(x) = \int f(x - y)g(y)dy$$

Proof.

$$F(x) = \int_{-\infty}^x f(x) dx$$

Since everything is nonnegative, from Fubini's theorem, we get

$$\begin{aligned}
 P(X + Y \leq z) &= \int \int_{-\infty}^{z-y} f(x) \, dx \, dG(y) \\
 &= \int \int_{-\infty}^z f(x-y) \, dx \, dG(y) \\
 &= \int_{-\infty}^z \int f(x-y) \, dG(y) \, dx \\
 &= \int_{-\infty}^z h(x) \, dx
 \end{aligned}$$

which shows that $X + Y$ has density $h(x)$.

When Y has density g , then

$$\int f(x-y) \, dG(y) = \int f(x-y) g(y) \, dy$$

□

Example 2.1.19 ► The gamma density

The **gamma density** with parameters α and λ is given by

$$f(x) = \begin{cases} \lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha) & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$.

Theorem 2.1.20

If $X = \text{gamma}(\alpha, \lambda)$ and $Y = \text{gamma}(\beta, \lambda)$ are independent, then $X + Y$ is $\text{gamma}(\alpha + \beta, \lambda)$. Consequently if $X_1, \dots, X_n$ are independent exponential (λ) r.v's, then $X_1 + \dots + X_n$ has a $\text{gamma}(n, \lambda)$ distribution.

Example 2.1.21 ► Normal distribution

In Example 1.6.2, we introduced the normal density with mean μ and

variance a ,

$$(2\pi a)^{-1/2} \exp(-(x - \mu)^2/2a).$$

Theorem 2.1.22

If $X = \text{normal}(\mu, a)$ and $Y = \text{normal}(\nu, b)$ are independent, then $X + Y = \text{normal}(\mu + \nu, a + b)$.

2.2 Weak Laws of Large Numbers

Index

Definitions

1.1.1	Probability Spaces . .	2
1.1.5	Semialgebra	3
1.1.7	Algebra	3
1.1.9	measure on an algebra	4
1.1.10	σ -finite	4
1.2.1		6
1.2.3	distribution	6
1.2.6	Equal in Distribution	8
1.3.1		9
1.3.3		10
1.4.1	Expected Value	12
1.4.9		13
2.1.1	Independence	15
2.1.3		16
2.1.6	Independence be- tween conlections . .	18
2.1.8	π, λ - systems	18

Examples

1.1.2	Discrete probability spaces	2
1.1.3	Measures on the real line.	2
1.1.6		3
1.2.2		6
1.2.7	Uniform distribution on (0,1)	9
1.2.8	Exponential distri- bution with rate λ . .	9
1.2.9	Standard normal dis- tribution	9
1.3.4		10
1.3.5		10

1.4.11		14
2.1.5		17
2.1.19	The gamma density .	23
2.1.21	Normal distribution .	23

Exercises

1.1.13		5
1.3.10		11
2.1.2		15
2.1.4		16

Propositions

1.4.10		14
--------	-----------	----

Lemmas

1.1.8		3
1.1.12		4
2.1.7		18

Theorems

1.1.4		2
1.1.11		4
1.2.4		7
1.2.5		7
1.2.10	For $x > 0$	9
1.3.2		10
1.3.6		10
1.3.7		10
1.3.8		10
1.3.9		11
1.4.2		12

1.4.3	Jensen	12	2.1.11		20
1.4.4	Holder	12	2.1.12		20
1.4.5	Fatou's lemma	13	2.1.13		20
1.4.6	Monotone conver- gence theorem	13	2.1.14		21
1.4.7	Dominated conver- gence theorem	13	2.1.15		21
1.4.8	Change of variables formula	13	2.1.16		21
2.1.9	$\pi - \lambda$ <i>Theorem</i>	18	2.1.17		21
2.1.10		18	2.1.18		22
			2.1.20		23
			2.1.22		24